MIL-P-50535 (MU)
7 December 1970
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

-

PROJECTILE, 155MM, HE: M449A1,

BODY ASSEMBLY FOR

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1. SCOPE

1.1 This specification covers the metal parts for one type of body assembly designated as Body Assembly, 155MM, HE: M449A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-S-20426	-	Shells, Base Ejection and Illuminating type (other than Mortar Shell), Metal parts for

STANDARDS

FSC: 1320

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MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

F-7549275	-	Pallet for 155MM Projectile
F-8875855	-	Body Assembly

PUBLICATIONS

ARMY

ET8875855	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 8-69	-	Tension Testing of Metallic Materials
ASTM E 10-66	-	Brinell Hardness of Metallic Materials

(Applications for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Body assemblies.-The body assemblies shall comply with all requirements specified on Drawing (Dwg.) F-8875855 all associated drawings, and with all requirements specified in applicable specifications.

3.3 Rotating band.

3.3.1 Band seat.-Prior to banding, the band seat shall be clean and free of oil, grease, dirt, rust and other foreig material. Shot or sand blasting is not permitted for cleanir band seats after knurling.

3.3.2 Band application.-The rotating band blank shall b carefully pressed into the band seat after the final heat tre ment of the projectile body so as to fit tightly with no dis- tortion of the adjacent walls and comply with the applicable drawing requirements.

3.3.3 Band clearance (tightness).-The clearance betweer the band and the body band seat in a radial direction shall r exceed 0.006 inch when tested as specified in 4.4.2.

3.3.4 Band sidewall contact (band fill).-The rotating band shall make contact with the sidewalls of the band seat f a full 360 degrees and a minimum of 75 percent of the height the sidewalls when tested as specified in 4.4.3.

3.4 Tension tests of rotating band.-The rotating band, after removal from the body assembly, shall comply with the tension test requirements specified in MIL-S-20426 when teste as specified in 4.4.4.

3.5 Bend test of rotating band.-The rotating band, afte removal from the body assembly, shall comply with the bend te requirements specified in MIL-S-20426 when tested as specifie in 4.4.5.

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3.6 Workmanship.-Requirements for workmanship shall be in accordance with the applicable drawings, MIL-A-2550, referenced specifications and the following:

3.6.1 Burr.-No assembly shall have a burr which might interfere with assembly or functioning of the item, or which might be injurious to personnel handling the item.

3.6.2 Surface preparation.-Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.6.3 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing. The body cavity shall be free of scale, fins, deep seams, die scratches, pits or other surface defects.

3.6.4 Metal defects.-All bodies shall be free from voids, cracks, seams, splits, laps, cold shuts, pipe, porosity, inclusions and other similar defects.

3.6.5 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of MIL-A-2550.

3.6.6 Rotating Band.-The machined rotating band shall not be nicked or otherwise damaged in subsequent handling prior to shipment.

3.6.7 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. All marking and stamping shall be neat and sharply defined.

3.6.8 Foreign Matter.-No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Discoloration not due to corrosive action, will be permitted.

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3.6.9 Obturator band.-The obturator bands shall be uniform in color, free of cracks, burn marks, contamination and shall have a smooth finish. Surface irregularities resulting from cutting off sprues and gates shall be removed by grinding and buffing until no protrusions exist.

3.6.10 Heat treatment.-All heat treatment (including stress relief) procedures shall be controlled so as to produce in each lot of components the mechanical properties specified on the applicable component drawings and in applicable specifications. (See 4.2.4).

3.7 Proving ground ballistic tests (first article sample only).-A first article sample (See 4.2.1.g.) shall be subjected to the proving ground ballistic tests specified in 4.2.2.1 and shall show no evidence of metal parts failure when tested as specified in 4.4.10 (See 6.1). The following conditions shall be considered as evidence of failure:

(1) Loss, separation, breakup, or fracture of metal parts in gun bore or during flight.
(2) Shear or loss of rotating band.
(3) Metal parts failure during expulsion phase.

3.8 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract (See 6.4).

3.9 First article testing.

3.9.1 Provisions.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.9.2 Heat treatment (mechanical and physical properties).-The contractor shall determine the procedures relating to cold-working and heat treatment which will produce parts meeting the required mechanical and physical properties. These procedures shall be documented and submitted as specified in 4.2.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Ten (10) unbanded, unpainted bodies
b. Ten (10) unpainted body assemblies
c. Five (5) painted, banded body assemblies
d. Twenty-five (25) rotating bands
e. Twenty-five (25) obturator bands
f. Three (3) slugs representing the steel blooms or billets with hot-sheared ends which were used in producing the body assemblies. (Applicable when this process is used)
g. Twenty (20) painted body assemblies (See 4.2.2.1)

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. The

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sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings. In addition, the following tests shall be performed:

4.2.2.1 Proving ground ballistic tests (See 3.7 and 6.6). The twenty (20) painted body assemblies from 4.2.1 (g) shall be forwarded to the location designated by the Contracting Officer for loading. The loaded assemblies shall then be subjected to the following ballistic tests. Testing shall be conducted at a Government-owned proving ground using the methods and procedures specified in 4.4.10.

4.2.2.1.1 Five (5) of the loaded assemblies shall be fired at Zone 7. (See 4.4.11.1)

4.2.2.1.2 Five (5) of the loaded assemblies shall be subjected to a five-foot drop test and fired at Zone 7. (See 4.4.11.2)

4.2.2.1.3 Ten (10) of the loaded assemblies shall be fired at Zone 8X. (See 4.4.11.3)

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted (including successful completion of all ballistic tests), the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

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4.2.4 Mechanical properties.-Documentation detailing all equipment, operations, and procedures to be used during the term of this contract, and affecting one or more of the mechanical properties, shall be submitted as part of first article inspection. Changes in the established procedures or equipment during the course of the contract shall be documented and submitted, with test data, as specified in 6.4. Documentation shall include actual chemical and metallurgical analyses of material, including wrought materials, (hardness, grain size and grain structure with photographs). The finished items shall also provide data for tensile strength, yield strength, percent elongation, percent reduction in area and charpy v-notch impact tests. Documentation covering normalizing, tempering, stress-relieving and annealing, when any of these operations are used, shall provide the shortest and longest time at the temperature for each item, size and type of furnace or oven. When quenching is employed, documentation shall indicate the quenching medium, its temperature, method of presentation of items into the medium and length of time in the medium. This documentation, with actual test data, shall be signed by a responsible agent, of the prime contractor and submitted to the technical agency specified in 6.4.

4.3 Inspection provisions.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardles s of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain band blanks of one lot interfix number from one manufacturer. (See 4.3.1.2).

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4.3.1.1 Heat treatment lot.-A heat treatment lot (includes stress-relief) shall consist of bodies from one mill-heat of steel that have been heat-treated in the same furnace at the same temperature and for the same length of time. Test samples shall be selected after final heat treatment or stress-relief. Where heat treatment is done in a continuously operated, conveyor-type furnace, the identification maintained shall be that code number assigned the particular mill-heat of steel. Where heat treatment or stress-relief is performed in batch-type furnaces, additional identification control must be maintained that directly relates to that portion of the mill-heat processed as a batch. The method of control must meet with the approval of the Government. Components in heat treatment lots shall remain grouped together for identifying purposes until mechanical properties tests have been completed. Components from heat-treatment lots which comply with the applicable mechanical properties requirements may be subsequently re-grouped into projectile lots.

4.3.1.2 Obturator band lot.-The obturator band lot shall be that quantity from the same batch of resin processed during one continuous cycle, at one time, at the same temperature and same pressure.

4.3.1.3 Lot identification.-Heat treatment lots and obturator band lots shall be suitably identified using an approved number or letter system. Identification by color shall not be used.

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4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspections for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Body (prior to banding) (See Dwg F-8875856 covering a detail of Dwg F-8875855).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Diameter of rotating band seat	Gage	01001
102.	Width of rotating band seat	Gage	01002
103.	Position of rotating band seat	Gage	01003
104.	Diameter of obturator band seat	Gage	01004
105.	Width of obturator band seat	Gage	01005
106.	Distance between band seats	Gage	01006
107.	Concentricity of base plug cavity diameter with bourrelet diameter	Gage	01007
108.	Depth of base plug cavity	Gage	01008
109.	Diameter of base plug cavity	Gage	01009
110.	Scoring of rotating band seat missing	Visual	01010
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship in band seats (See 3.3.1)	Visual	01011

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4.3.2.2 Body Assembly (prior to painting and application of interior coating) (See Dwg F-8875855).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 2.50 Percent		
101.	Concentricity of rotating band with bourrelet	Gage	02001
102.	Concentricity of pitch diameter of nose thread with bourrelet diameter including perpendicularity with nose face	Gage	02002
103.	Concentricity of boattail angle with bourrelet diameter	Gage	02003
104.	Concentricity of inside diameter with bourrelet diameter (including straightness)(begin one-quarter inch from pusher plate seat and measure every two inches)	Gage	02004
105.	Pitch diameter of nose thread	Gage	02005
106.	Minor diameter of nose thread, max	Gage	02006
107.	Diameter of bourrelet, min (including portion between two bands)	Gage	02007
108.	Second largest outside diameter	Gage	02008
109.	Distance from nose face to forward end of bourrelet	Gage	02009
110.	Distance from nose face to rear end of bourrelet	Gage	02010
111.	Inside diameter of large cavity	Gage	02011
112.	Depth of large cavity, including straightness	Gage	02012
113.	Distance from base end to larger internal datum diameter near bottom of large cavity	Gage	02013
114.	Wall thickness of second cavity (charge cavity), min	Gage	02014
115.	Diameter of flat portion of wall bottom of charge cavity, max.	Gage	02015
116.	Wall thickness in indicated area in bottom of large cavity, min	Gage	02016
117.	Total length	Gage	02017
118.	Outside diameter at nose	Gage	02018

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119.	Distance from base end to rotating band	Gage	02019
120.	Width of rotating band	Gage	02020
121.	Perpendicularity of rear face with bourrelet diameter	Gage	02021
122.	Location of larger external datum from base end	Gage	02022
123.	Location of smaller external datum from base end	Gage	02023
124.	Distance between larger internal datum diameter and smaller internal datum diameter (bottom of large cavity)	Gage	02024
125.	Wall thickness variation, max, in indicated area	Gage	02025
126.	Finish improper in bottom of large cavity before forward plate assembly seat (See 3.6.3)	Gage	02026
127.	Concentricity of outside diameter on ogive with bourrelet at indicated distance from nose face	Gage	02027
128.	Band seat width not completely filled	Visual	02028
Minor	AQL 1.50 Percent		
201.	Excessive burr on thread	Visual	02029
202.	Base plug chamfer improper	Visual	02030
203.	Finish improper (excluding area covered by 126) (See 3.6.3)	Visual	02031
204.	Evidence of poor workmanship (See 3.6.8)	Visual	02032

4.3.2.3 Body Assembly, prior to assembling lifting plug (See Dwg F-8875855).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Weight	Balance	03001
102.	VCI paper missing from body cavity	Visual	03002

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Minor		AQL 1.00 Percent	
201.	Lubricant missing from lifting plug threads	Visual	03003
202.	Internal coating missing or inadequate	Visual	03004

4.3.2.4 Obturator band (after moisture-conditioning) (See Dwg C-8875844covering a detail of Dwg F-8875855).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Inside diameter	Gage	04001
102.	Width	Gage	04002
103.	Location of holes	Gage	04003
104.	Thickness	Gage	04004
Minor	AQL 1.50 Percent		
201.	Chamfer missing or improper	Visual	04005
202.	Flash	Visual	04006
203.	Evidence of poor workmanship (See 3.6.1)	Visual	04007

4.3.2.5 Body assembly (See Dwg F-8875855).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Diameter of bourrelet, max (including portion between bands)	Gage	05001
102.	Diameter of rotating band, max	Gage	05002
103.	Diameter of obturator band, min	Gage	05003
104.	Angle incorrect on rotating band	Gage	05004
105.	Rotating band missing or damaged	Visual	05005
106.	Grommet missing	Visual	05006
107.	Lifting plug missing	Visual	05007
Minor	AQL 1.50 Percent		
201.	Protective coating with bare spot except slight scratches	Visual	05008
202.	Marking misleading or unidentifiable	Visual	05009
203.	Evidence of poor workmanship (See 3.6)	Visual	05010

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4.3.2.6 Pallet (loaded) (See Dwg F-7549275).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Strap width incorrect	Visual	06001
102.	Strapping improperly assembled	Visual	06002
103.	Connector missing or improperly assembled	Visual	06003
104.	Body assembly missing	Visual	06004
105.	Board broken or split	Visual	06005
106.	Markings for FSN or descriptive nomenclature missing, incorrect or unidentifiable	Visual	06006
107.	Body assembly not seated on VCI paper	Visual	06007
Minor	AQL 1.00 Percent		
201.	Marking, other than above, missing, incorrect or unidentifiable	Visual	06008
202.	Evidence of poor workmanship (See 3.6.7 and 3.6.8)	Visual	06009

4.3.3 Testing.

4.3.3.1 Surface defects of body assembly (prior to phosphate coating and prior to painting) (See Dwg F-8875855) Major defect, Code No. 07001.-This inspection shall be performed 100%. Each body assembly shall be inspected visually for cracks, voids, seams and laminations. If any body assembly contains any of these defects it shall be classed defective and removed from the lot. The examination shall be performed as specified in 4.4.1.

4.3.3.2 Rotating band.

4.3.3.2.1 Rotating band clearance (tightness) (See 3.3.3) Major defect, Code No. 08001.-One (1) body assembly shall be selected from each eight (8) hour's production, or fraction thereof, from each banding machine in operation and subjected to this test. If any body assembly fails to comply with the applicable requirement, it shall be classed defective and the quantity of bodies banded since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

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4.3.3.2.2 Band sidewall contact (See 3.3.4) Major defect, Code No. 09001).-One (1) body assembly shall be selected from each eight (8) hour's production, or fraction thereof, from each banding machine in operation and subjected to this test. If any body assembly fails to comply with the applicable requirement, it shall be classed defective and the quantity of bodies banded since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.2.3 Tension tests of rotating band (See Dwg. F-8875855 and MIL-S-20426) Major defect, Code No. 10001.-One (1) body assembly shall be selected from each eight (8) hour's production or fraction thereof, from each banding machine in operation and the band subjected to these tests. If any band fails to comply with the mechanical properties requirements (tensile strength and elongation) specified on the applicable drawing, it shall be classed defective and the quantity of bodies banded on that machine since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4. The body assemblies selected for rotating band sidewall contact may be used for this test.

4.3.3.2.3.1 Retest.-A retest will be permitted at the request of the contractor. A random sample of five (5) body assemblies shall be selected from the quantity represented by each failing band and tested as specified in 4.4.4. If any rotating band fails to comply with the applicable requirements, that quantity of bodies represented by the original failing sample shall be rejected.

4.3.3.2.4 Bend test of rotating band (See Dwg F-8875855 and MIL-S-20426) Major defect, Code No. 11001.-One (1) body assembly shall be selected from each eight (8) hour's production, or fraction thereof, from each banding machine in operation and the band subjected to this test. If any band fails to comply with the bend test requirements specified, it shall be classed defective and the quantity of bodies banded since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4. The body assemblies selected for rotating band clearance or tensile strength may be used for this test.

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4.3.3.2.4.1 Retest.-A retest will be permitted at the request of the contractor. A random sample of five (5) body assemblies shall be selected from the quantity represented by each failing band and tested as specified in 4.4.5. If any rotating band fails to comply with the applicable requirements, the quantity of bodies represented by the original failing sample shall be rejected.

4.3.3.3 Tension tests of body (prior to machining and after completion of all heat treatment operations, including stress-relieving (See Dwg F-8875856) Major Defect, Code No. 12001). Sample bodies shall be selected from each mill heat of steel (See 4.3.1.1) in accordance with Table I and specimens cut therefrom. If any test specimen fails to comply with the mechanical properties requirements specified on the applicable drawing, that quantity of bodies represented by the sample specimen shall be rejected. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4.

TABLE I

No. of Bodies in Mill Heat Lot	Sample Size
50-200	2
201-300	3
301-400	4
401-500	5
501-600	6
601-700	7
701-800	8
801-900	9
Over 900	10

If more than one heat treatment (including stress-relief) batch is included in the mill-heat of steel, or when the heat of steel is processed in two (2) or more batch-type furnaces, the sample bodies shall be selected in such a manner as to be representative of the greatest number of batches possible within the mill heat. When heat treatment is performed in a continuously operated, conveyor-type furnace, the samples shall be selected in a sequence that will be representative, to the maximum extent, of the entire mill heat lot in process.

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4.3.3.4 Hardness of body (prior to machining and after completion of all heat treatment operations) (See Dwg F-8875856) Major Defect, Code No. 13001 (when applicable).-The bodies selected to take the specimens for the tensile tests specified in 4.3.3.3 shall be used as samples for this test. The test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4. If the average value for hardness is not within the hardness limits specified on the applicable drawing, that quantity of bodies represented by the sample shall be rejected.

4.3.3.5 Moisture content of obturator band (See Dwg C-8875844) Major Defect, Code No. 14001.-Ten (10) obturator bands shall be selected from each lot and subjected to this test. The test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4. If two (2) or more bands fail to comply with the applicable drawing requirements, the lot shall be rejected. If one band fails to comply with the applicable drawing requirements, the lot shall be retested.

4.3.3.5.1 Retest.-An additional sample of ten (10) obturator bands shall be selected from the lot for this test. If one (1) or more bands in the second sample fail to comply with the applicable drawing requirements, the lot shall be rejected.

4.3.3.6 Tension test of obturator band (See Dwg C-8875844 and 4.3.1.2) Major Defect, Code No. 15001.-Ten (10) obturator bands shall be selected from each lot and subjected to this test. The test shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4. If two (2) or more bands fail to comply with the applicable drawing requirements, the lot shall be rejected. If one band fails to comply with the applicable drawing requirements, the lot shall be retested. The bands selected for the moisture content determination (See 4.3.3.5) may be used for this test.

4.3.3.6.1 Retest.-An additional sample of ten (10) obturator bands shall be selected from the lot for this test. If one (1) or more bands fail to comply with the applicable drawing requirements, the lot shall be rejected.

4.3.3.7 Elongation of obturator band (See Dwg C-8875844 and 4.3.1.2) Major Defect, Code No. 16001.-Five (5) test coupons which are processed with each lot of obturator bands, and conditioned as specified on the applicable drawing, shall be used for this test. The sample coupons shall have the size and shape as

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indicated on the specified ASTM. The thickness of the sample coupons shall be equivalent to the obturator band thickness as processed. If the average percent of elongation of the five (5) sample coupons does not comply with the applicable requirements, the lot shall be rejected. The test shall be performed as specified in 4.4.10 using equipment in accordance with 4.3.4.

4.3.3.8 Tool control for radius at bottom of large cavity at forward end (See Dwg F-8875856 - note 4) Major Defect, Code No. 17001.-In place of the normal sampling associated with the classifications of defects, and with the approval of the Contracting Officer, the following procedure shall apply: At the start of production, after every four (4) hours of use, and whenever a change or modification of punching tools is made during production, a tool radius measurement shall be obtained. Determination of tool radius through the use of wax impression is permissible. The contractor's tolerance limits for control of tool radius shall be approved by the procuring activity. When a measurement falls outside the contractor's approved tolerance limits, twenty-five (25) body assemblies, from the latter portion of production, from that tool, since the last acceptable measurement was taken, shall be measured for nose end radius. The use of wax impressions is a permissible means of obtaining these measurements. If any nose end radius measurement, taken in this manner, falls outside the tolerance required by the component drawing, the body assemblies produced from that tool since the last acceptable measurement was taken, shall be inspected one hundred percent by the contractor to remove any bodies which are outside drawing tolerance.

In addition, measurements of nose end radius shall be taken after forming of radius from five (5) body assemblies from each eight (8) hour's production, or portion thereof. The use of wax impressions is a permissible means of obtaining these measurements. If any measurement falls outside the drawing tolerance, the body assemblies produced since the last satisfactory measurement was obtained shall be inspected one hundred percent by the contractor to remove any body assembly which does not comply with the applicable drawing requirement.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8875855 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

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4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Surface defects of body assembly.-Inspection for cracks, voids, seams and laminations shall be performed on the interior and exterior of the body, prior to application of any coating or painting. Interior inspection shall be conducted by means of a light and a mirror or any other method approved by the Contracting Officer.

4.4.2 Rotating band clearance.-Prior to removal, the diameter of the finished band shall be measured on three (3) diameters, 60° apart, and on two (2) points along the cylindrical portion of its length near the front and near the rear edge. This comprises six (6) diametral measurements. The points of measurement shall be marked on the bands as well as the orientation of the band on the body. The band shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of twelve (12) measurements), and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat, the sum of the corresponding two (2) thicknesses of the band and subtract the result from the corresponding exterior diameter of the band previously determined. The result will indicate the total diametral clearance between the band and its seat. If the band is well seated, a negative clearance, or interference, may be indicated on some of the measurements or even on the average of all measurements. When waved band ribs or irregular band profile interfere with ease of measurement, the manufacturer is authorized to machine the outside of the band in order to obtain the adequate cylindrical surface. The diameter of the machined band must exceed the bourrelet diameter.

4.4.3 Band sidewall contact.-The sample selected in accordance with 4.3.3.2.2 shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward and rear of rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other half shall be sectioned into two parts, thus exposing four surfaces each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the specified requirements.

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4.4.4 Tension tests of rotating band.-The band shall be carefully removed from the body and carefully flattened without hammering. The test specimen shall be cut along the direction of the flattened length and prepared in such a manner as to hold to the minimum any additional cold working. Tension test specimens shall be machined to the form and dimensions specified in MIL-S-20426. The specimens shall be tested as specified in MIL-S-20426. Observation shall be made for tensile strength and percentage of elongation.

4.4.5 Bend test of rotating band.-The band, or section thereof, of full width and thickness except as reduced by authorized surface cut, shall be bent cold by pressure, without hammering, in the direction of curvature of the band, until it tightly encloses 180 degrees of a rod equal in diameter to the thickness of the band test specimen (within plus or minus 0.010 inch). The band shall be observed for cracks resulting from the test.

4.4.6 Tension tests of body.-Prior to machining of the body, and following completion of all heat treatment operations, including stress-relief, one (1) tension specimen conforming to either ASTM E 8-69, Fig. 6, Sheet-type, or Figures 1 and 2 of this specification, shall be taken longitudinally from the wall of each sample body. The test specimen shall be taken in such a position that the gage length will include the section of the wall at the location of the rotating band, and in such a manner that the test specimen is from within the contours of the finished projectile. The tests shall be conducted in accordance with ASTM E 8-69 (part 31). Yield strength shall be determined by the offset method, The offset shall be 0.1 percent (0.001 inch per inch of gage length). Identity of each heat treated lot shall be maintained until minimum mechanical properties (See 4.3.1.1) have been verified by the inspector.

4.4.7 Hardness of body.-Six (6) standard Brinell hardness readings from each zone shall be taken approximately $\frac{1}{4}$ inch apart and 1/32 inch from the inside and outside wall in a manner to form two (2) horizontal rows (three in each row). The average value for the six (6) hardness readings shall be determined. No readings shall be deleted from the average unless it can be determined that they are the result of improper operation of the Brinell equipment or improper preparation of the test sample. The test shall be performed in accordance with ASTM E 10-66 (Part 31).

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4.4.8 Moisture content of obturator band.-The obturator bands, after weighing on an analytic balance, shall be heated in an air-circulating oven for 24 hours at a temperature of 105 degrees centigrade (^{o}C). The bands shall then be cooled to room temperature in a desiccator and reweighed. The bands shall then be heated for one hour at 105°C, cooled and again weighed. The procedure for heating of one hour, cooling and weighing will be continued until the bands reach a constant weight. The difference between the initial weight and the final constant weight shall be taken as the amount of moisture originally present in the band. The moisture content (%) is calculated as follows:

$$\text{Moisture Content (\%)} = \frac{\text{Initial weight - final constant weight X 100}}{\text{Initial weight}}$$

4.4.9 Tension test of obturator band.-The obturator band shall be assembled in a fixture and tested as specified on the Equipment List. The tensile strength shall be obtained by dividing the load recorded at failure by two times the cross sectional area of the band excluding the cross sectional area of the holes. With the band and fixture in place, the tensile testing machine will be operated at a constant rate of 2.0 inches per minute until the band breaks.

4.4.10 Elongation of obturator band.-The obturator band shall be tested as specified in the ASTM referenced on the applicable drawing.

4.4.11 Proving Ground ballistic tests (See 3.7 and 6.7). The tests specified in 4.2.2.1 shall be conducted at a Government-owned proving ground. A suitable recovery area shall be prepared prior to firing any rounds. The area shall minimize the effects of impact on the body assembly and permit recovery of the body assemblies intact for examination. All rounds shall be suitably marked for identification purposes.

All rounds shall be fired from a weapon for which they are standard, with the fuze assembly (with expelling charge assembly, see Dwg. CP-133386) set to achieve a minimum burst height of fifteen hundred (1500) feet. Photographic equipment shall be positioned near the gun muzzle to record the flight of each round. Examination of photographs and recovered body assemblies shall be made for any evidence of failure of the metal parts in order to determine compliance of each round with the requirements of 3.7.

Those rounds which are to be inert-loaded may contain wax, borax or other approved inert material in place of the explosive.

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All test results and other pertinent information shall be recorded and shall include the following:

a. Evidence of failure to comply with any of the requirements in 3.7.
b. Rupturing or cracking of ogive in wall area at bottom of large cavity in body.
c. Probable cause of defect (impact, ballistically caused, manufacturing defect, etc.)

4.4.11.1 Zone 7.-The five (5) body assemblies to be tested at Zone 7 shall be inert-loaded. The cargo may contain live primers. All rounds shall be fired at ambient temperature using the appropriate Zone 7 propelling charge increments.

4.4.11.2 Five-foot drop and Zone 7 firing.-The five (5) body assemblies shall be inert-loaded. The cargo may contain live primers. Rounds, with lifting plug assembled, shall have a five-foot drop test performed prior to firing. The projectiles shall be dropped five feet (lowest point of the projectile to point of impact) onto a steel plate which is solidly supported by concrete. The orientation of the projectile shall be at an axis 45 degrees from the vertical, with the base down. The projectile will be attached by means of a chain or cable at the lifting plug to prevent the projectile from tumbling after impact. The armor plate upon which impact occurs shall have a minimum thickness of 3 inches, a Brinnel hardness of at least 207, and shall be solidly supported in a horizontal plane over its entire bearing area by a minimum thickness of 24 inches of concrete. The steel plate shall have a smooth and flat surface. Upon completion of the drop test, the projectile shall be examined for evidence of separation, loosening, or fracture of any body assembly or base plug. Projectiles showing no evidence of these defects, as determined by the Proof Director, shall then have the fuze assembled. The rounds shall be fired at ambient temperature using the appropriate Zone 7 propelling charge increments.

4.4.11.3 Zone 8X.-The ten (10) body assemblies to be tested at Zone 8X shall be HE-loaded. Prior to firing, the rounds shall be temperature-conditioned for not less than 24 hours at the temperature specified $\pm 5^{\circ}$:

Five (5) of the rounds shall be conditioned to -65°F.
Five (5) of the rounds shall be conditioned to $+165^{\circ}$F.

The rounds shall be fired using a propelling charge adjusted to give 112 percent of the rated pressure of the propelling charge. (The rated pressure appears on the data card of each propelling charge lot.)

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (Level C).-See 5.2

5.2 Packing (Level C).-Eight (8) body assemblies shall be packed in accordance with Dwg F-7549275.

5.3 Marking.-Marking shall be in accordance with Dwg. F-7549275.

5.4 Shipping.-When assemblies from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment

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coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6 3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A , Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

6.5 Supersession data.-This military specification includes the requirements of, and supersedes, Purchase Specification AEI-204B, dated 3 January 1968.

6.6 Base plugs.-The Contracting Officer shall arrange for the shipment of twenty (20) base plugs (Dwg. C-8875854) to the load plant to be assembled to the loaded body assemblies.

6.7 Results of proving ground ballistic tests.-A copy of the ballistic test results and any other requested information relating to these tests shall be forwarded to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No: 1320

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TENSILE TEST SPECIMEN

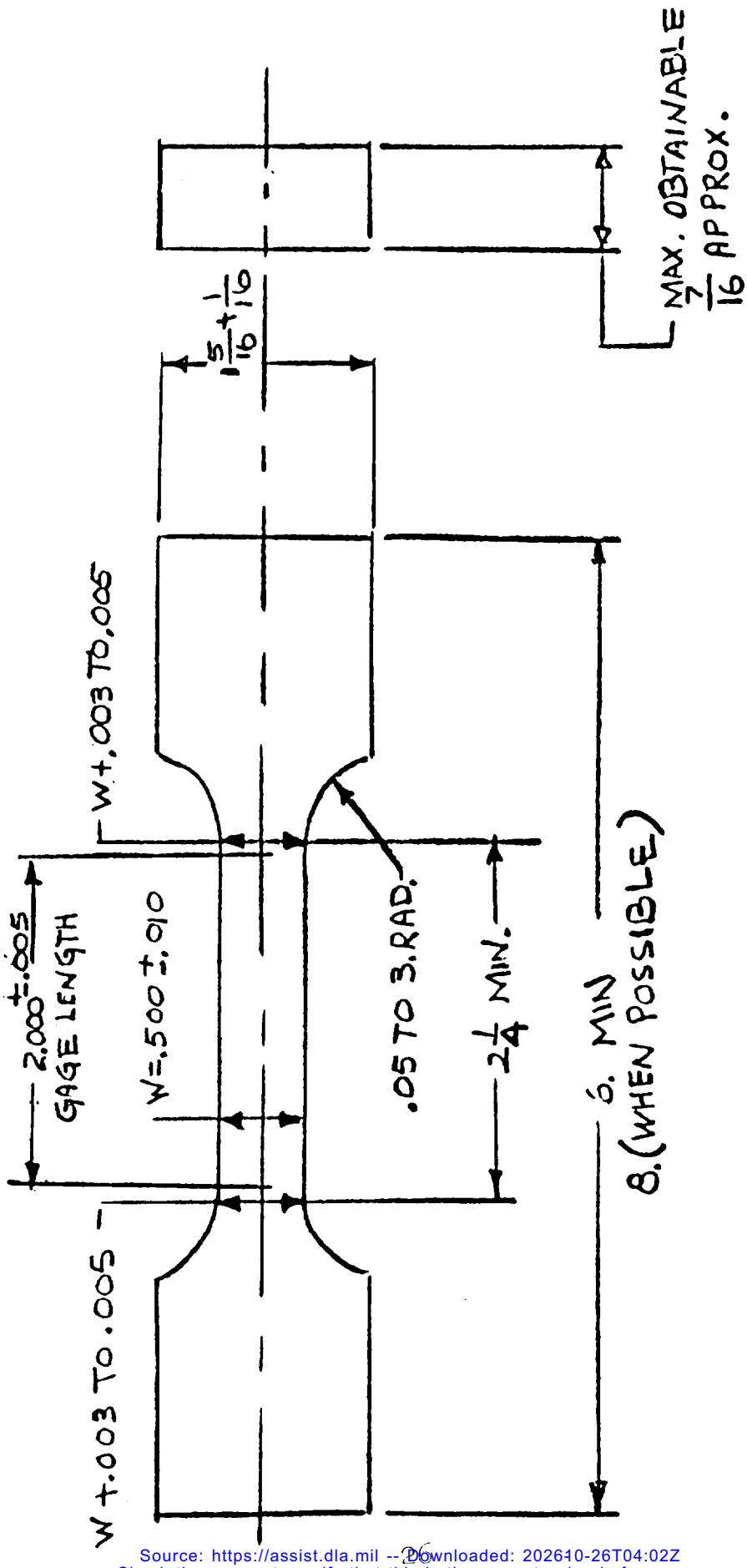


FIGURE 1

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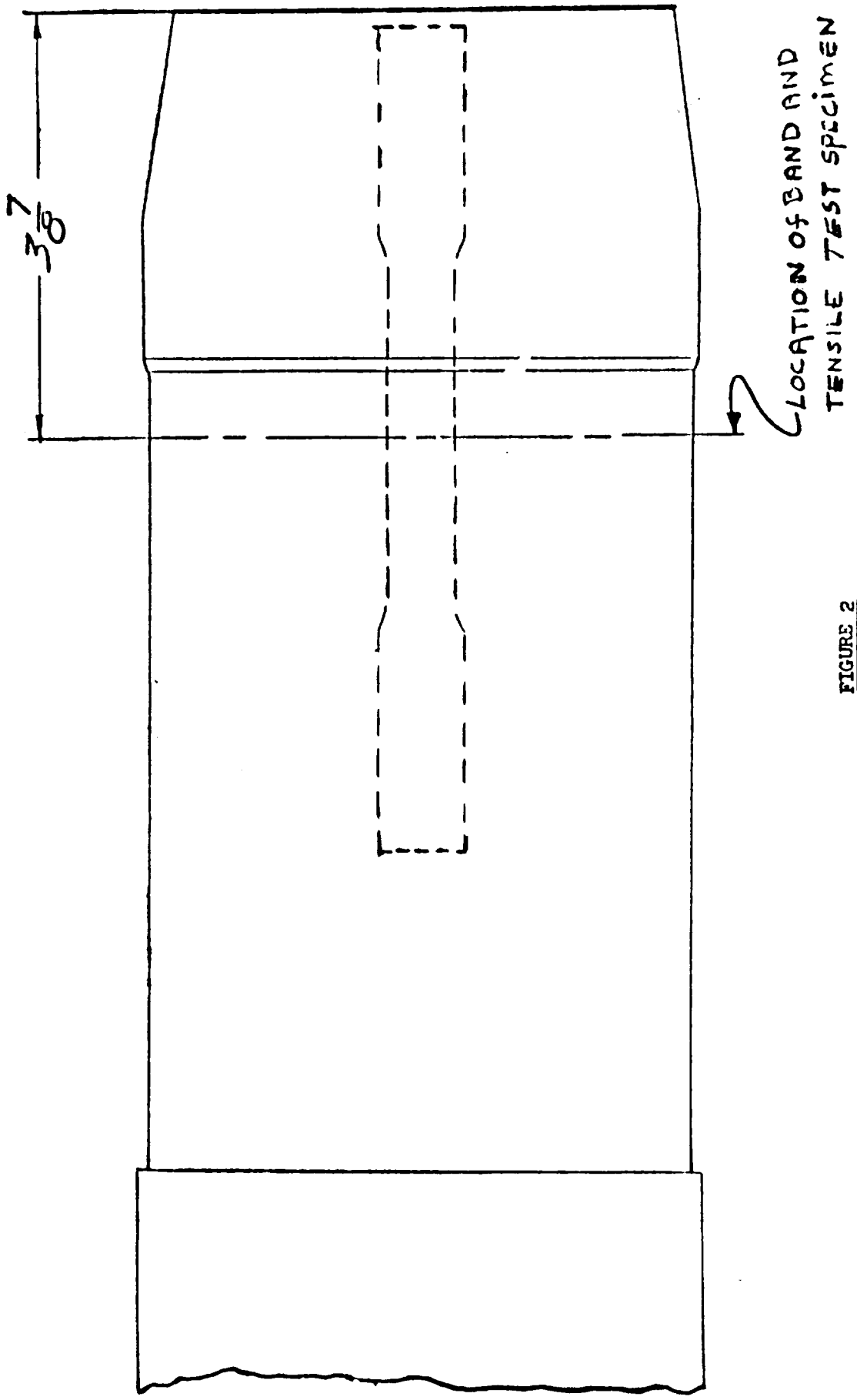


FIGURE 2

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